

Queries in DBMS

What is a Query?

- A query is a request to retrieve or manipulate data from a database.
 - Queries are written using Structured Query Language.
 - Queries help users access required information from tables.
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Definition

Query:

A command used to access, insert, update, delete, or retrieve data from a database.

Importance of Queries

1. Retrieve required data
 2. Filter records
 3. Update information
 4. Delete unwanted data
 5. Generate reports
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Types of Queries in DBMS

1. Select Query
 2. Insert Query
 3. Update Query
 4. Delete Query
 5. Nested Query
 6. Aggregate Query
 7. Action Query
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1. SELECT Query

Definition

- Used to retrieve data from tables.
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Syntax

```
SELECT column_name  
FROM table_name;
```

Example

```
SELECT * FROM Student;
```

Explanation

- SELECT → retrieves data
 - * → all columns
 - Student → table name
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Output Example

Student_ID	Name	Course
101	Ravi	CSE
102	Anu	ECE

Important Points

6. Most commonly used query
 7. Retrieves rows from table
 8. Can retrieve specific columns
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2. INSERT Query

Definition

- Used to add new records into table.

Syntax

```
INSERT INTO table_name  
VALUES(values);
```

Example

```
INSERT INTO Student  
VALUES(103,'Kiran','CSE');
```

Important Points

- 9. Adds new rows
 - 10.Values should match column order
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3. UPDATE Query

Definition

- Used to modify existing data.
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Syntax

```
UPDATE table_name  
SET column_name=value  
WHERE condition;
```

Example

```
UPDATE Student  
SET Course='ECE'  
WHERE Student_ID=101;
```

Important Points

- 11.Changes existing records
 - 12.WHERE clause is important
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4. DELETE Query

Definition

- Used to remove records from table.
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Syntax

```
DELETE FROM table_name  
WHERE condition;
```

Example

```
DELETE FROM Student  
WHERE Student_ID=103;
```

Important Points

- 13.Removes selected rows
 - 14.WHERE clause avoids deleting all rows
-

5. Nested Query (Subquery)

Definition

- Query inside another query.
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Example

```
SELECT Name
FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);
```

Important Points

- 15. Inner query executes first
 - 16. Used for complex conditions
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6. Aggregate Queries

Definition

- Uses aggregate functions for calculations.
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Aggregate Functions

Function Purpose

COUNT() Counts rows

SUM() Finds total

AVG() Finds average

MAX() Finds highest value

MIN() Finds lowest value

Example

```
SELECT AVG(Marks)
FROM Student;
```

Important Points

- 17. Used for calculations
- 18. Works on groups of data

7. Action Queries

Definition

- Queries that perform actions on data.
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Types

19.INSERT

20.UPDATE

21.DELETE

Clauses Used in Queries

WHERE Clause

Definition

- Filters rows based on condition.

Example

```
SELECT * FROM Student  
WHERE Course='CSE';
```

ORDER BY Clause

Definition

- Sorts records.

Example

```
SELECT * FROM Student  
ORDER BY Name ASC;
```

GROUP BY Clause

Definition

- Groups rows with same values.

Example

```
SELECT Course, COUNT(*)  
FROM Student  
GROUP BY Course;
```

HAVING Clause

Definition

- Filters grouped records.

Example

```
SELECT Course, AVG(Marks)  
FROM Student  
GROUP BY Course  
HAVING AVG(Marks)>80;
```

JOIN Queries

Definition

- Combines data from multiple tables.

Example

```
SELECT Student.Name, Department.Dept_Name  
FROM Student  
JOIN Department  
ON Student.Dept_ID=Department.Dept_ID;
```

Types of Joins

- 22. INNER JOIN
 - 23. LEFT JOIN
 - 24. RIGHT JOIN
 - 25. FULL JOIN
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Advantages of Queries

- 26. Fast data retrieval
 - 27. Easy data management
 - 28. Supports filtering
 - 29. Helps generate reports
 - 30. Reduces manual work
 - 31. Improves analysis
 - 32. Supports decision making
 - 33. Efficient data handling
 - 34. Simplifies operations
 - 35. Works with large databases
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Additional Important Points

- 36. Queries are written using SQL.
- 37. SELECT retrieves data.
- 38. INSERT adds data.
- 39. UPDATE modifies data.
- 40. DELETE removes data.
- 41. WHERE filters rows.
- 42. ORDER BY sorts records.
- 43. GROUP BY creates groups.

44.HAVING filters grouped data.

45.JOIN combines tables.

46.Aggregate functions perform calculations.

47.Nested queries contain subqueries.

48.Queries improve database usability.

49.Queries are essential in DBMS.

50.SQL queries are widely used in applications.

Real-Time Example

College Database

Query

Display names of CSE students.

```
SELECT Name  
FROM Student  
WHERE Course='CSE';
```

Short Definitions for Exams

Query

A request to retrieve or manipulate data from a database.

Subquery

A query written inside another query.